

Advanced (Habanero)

- Q1.** Write the quadratic function $f(x) = x^2 + 4x + 4$ in vertex form.
- (A) $f(x) = (x + 2)^2$
(B) $f(x) = (x - 2)^2$
(C) $f(x) = (x + 1)^2$
(D) $f(x) = (x - 1)^2$
(E) $f(x) = x^2$
- Q2.** The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$. What does h represent?
- (A) the y -intercept
(B) the x -intercept
(C) the vertex's x -coordinate
(D) the vertex's y -coordinate
(E) the slope
- Q3.** Write the quadratic function $f(x) = x^2 - 7x + 12$ in intercept form.
- (A) $f(x) = (x - 3)(x - 4)$
(B) $f(x) = (x + 3)(x + 4)$
(C) $f(x) = (x - 2)(x - 5)$
(D) $f(x) = (x + 2)(x + 5)$
(E) $f(x) = (x - 1)(x - 6)$
- Q4.** What is the y -intercept of the quadratic function $f(x) = (x + 2)(x - 5)$?
- (A) -10
(B) 10
(C) -2
(D) 5
(E) 0
- Q5.** The quadratic function $f(x) = a(x - h)^2 + k$ has a vertex at (h, k) . If $h = 2$ and $k = -3$, what is the equation of the axis of symmetry?
- (A) $x = -2$
(B) $x = 2$
(C) $x = -1$
(D) $y = 2$
(E) $y = -3$
- Q6.** Write the quadratic function $f(x) = x^2 + 2x - 6$ in vertex form.
- (A) $f(x) = (x + 1)^2 - 7$
(B) $f(x) = (x - 1)^2 - 7$
(C) $f(x) = (x + 2)^2 - 10$
(D) $f(x) = (x - 2)^2 - 10$
(E) $f(x) = x^2$
- Q7.** What are the x -intercepts of the quadratic function $f(x) = (x + 1)(x - 4)$?
- (A) $x = -1, x = 4$
(B) $x = -4, x = 1$
(C) $x = 1, x = -4$
(D) $x = -2, x = 5$
(E) $x = 2, x = -5$
- Q8.** The quadratic function $f(x) = a(x - h)^2 + k$ has an axis of symmetry at $x = -3$. What is the value of h ?
- (A) -3
(B) 3
(C) 0
(D) 1
(E) -1

Open Ended Questions (FRQ)

- Q9.** Write the quadratic function $f(x) = x^2 - 9x + 20$ in both vertex and intercept form. Identify the vertex and the x -intercepts.
- Q10.** The quadratic function $f(x) = a(x - h)^2 + k$ has a vertex at $(2, -5)$. If the x -intercepts are $x = 1$ and $x = 3$, find the value of a . Show your work and explain your reasoning.

Chef's Tips

- Read each question carefully.
- Use the given equations and graphs to solve the problems.
- Show all work and justify your answers for free response questions.